However, none of the models achieve high overall scores (above 7.0), which may be influenced by the weaker experimental results from the previous step. This demonstrates that *experimental success is a critical determinant of overall research quality*.

Table 6: Evaluation results of three models averaged on ten tasks in paper writing. Best and worst scores are highlighted in green and red background, respectively.

Model	Consistency	Clarity	Completeness	Soundness	Overall
`o4-mini-high`	6.35 ± 1.13	7.25 ± 0.77	6.15 ± 0.77	5.05 ± 1.11	5.90 ± 0.97
`Gemini-2.5-Pro-Preview`	7.55 ± 0.91	8.05 ± 0.52	7.20 ± 0.77	6.05 ± 0.72	6.60 ± 0.86
`Claude-3.7-Sonnet`	7.40 ± 0.95	7.80 ± 0.61	6.80 ± 0.64	5.85 ± 0.89	6.50 ± 0.81

3.5 Results of End-to-End Evaluation

Current research agents struggle to generate robust scientific results. We conduct an end-to-end evaluation to compare three reasoning models within our MLR-Agent scaffold and benchmark them against AI Scientist V2 [39], with results summarized in Table 7. Our findings reveal two key points. Firstly, when using the MLR-Agent scaffold, the overall ratings for all three models fall below the 6.0 acceptance threshold. Notably, all models receive their lowest scores on *Soundness*, which underscores their shared difficulty in producing scientifically sound results. Secondly, in a direct model comparison, `o4-mini` is significantly outperformed by `Gemini` and `Claude`, whose performances are comparable. This trend is inversely correlated with cost. When using `Claude Code` as the coding agent, `o4-mini` is the most affordable, followed by `Gemini`. Consequently, `Gemini` emerges as the most cost-effective model for end-to-end autonomous research, considering its strong performance and moderate cost.

Table 7: End-to-end evaluation results over the ten tasks. AI Scientist V2 is powered by `o4-mini-high`. We use `o4-mini-medium` in `Codex` and use Gemini-2.5-Pro-Preview-05-06 in this experiment. When using `Claude Code` as the coding agent, the costs of `o4-mini-high`, `Gemini-2.5-Pro-Preview`, and `Claude-3.7-Sonnet` are \$1.15, \$1.24, and \$2.40, respectively.

Model	Clarity	Novelty	Soundness	Significance	Overall
AI Scientist V2					
`o4-mini-high`	6.55 ± 0.94	6.70 ± 0.48	3.70 ± 1.29	4.85 ± 1.08	4.25 ± 1.25
MLR-Agent					
`o4-mini-high + Codex`	6.45 ± 0.90	5.65 ± 0.60	2.90 ± 0.57	3.80 ± 0.65	3.10 ± 0.60
`Gemini-2.5-Pro-Preview + Gemini CLI`	8.30 ± 0.37	6.85 ± 0.34	4.15 ± 1.06	5.30 ± 0.91	4.60 ± 1.00
`Claude-3.7-Sonnet + Claude Code`	7.75 ± 0.34	7.10 ± 0.43	4.05 ± 1.11	5.50 ± 1.14	4.70 ± 1.22

AI Scientist V2 outperforms MLR-Agent, yet both share fundamental limitations. We further compare MLR-Agent with AI Scientist V2 in Table 7. Our comparison between MLR-Agent and AI Scientist V2, both using `o4-mini` as the backbone model, reveals that AI Scientist V2 consistently outperforms MLR-Agent across all review dimensions. However, neither agent meets the 6.0 acceptance threshold. A closer analysis of their detailed scores highlights three **key findings**:

- **Both agents excel at generating novel research outcome clearly.** Both agents demonstrate strong creativity, reflected in high scores for *Novelty*. They can propose original research questions crucial for scientific discovery and articulate these ideas clearly, as shown by their strong performance in the *Clarity* dimension.
- **Agents struggle with technical soundness.** A primary weakness is their inability to propose technically sound methods. This is evident from their markedly low scores in *Soundness*, which consistently fall below the acceptance threshold. This suggests that while the agents can formulate creative ideas, their technical execution is frequently flawed.
- **The agents' research demonstrates limited potential impact.** This limitation is reflected in their consistently low scores for *Significance*, suggesting their work often fails to address important research questions. This perceived lack of contribution, potentially compounded by reproducibility challenges, severely restricts the research's influence and uptake by the scientific community.

4 How Well Is MLR-Judge Aligned with Human Reviewers?

To evaluate whether our proposed LLM-based judge, **MLR-Judge**, can serve as a reliable proxy for human reviewers, we conduct a human evaluation study with ten domain experts who have served as reviewers for major machine learning conferences such as NeurIPS, ICLR, or ICML. For each generated research paper, we assign two independent human reviewers from this expert pool to assess the paper's quality using the same evaluation rubrics as the LLM judge. Specifically, all reviewers—whether human or MLR-Judge—follow the same review criteria, covering five dimensions: *Clarity*, *Novelty*, *Soundness*, *Significance*, and *Overall*. The detailed evaluation rubrics are provided in Table 28 and the process is introduced in Section E.

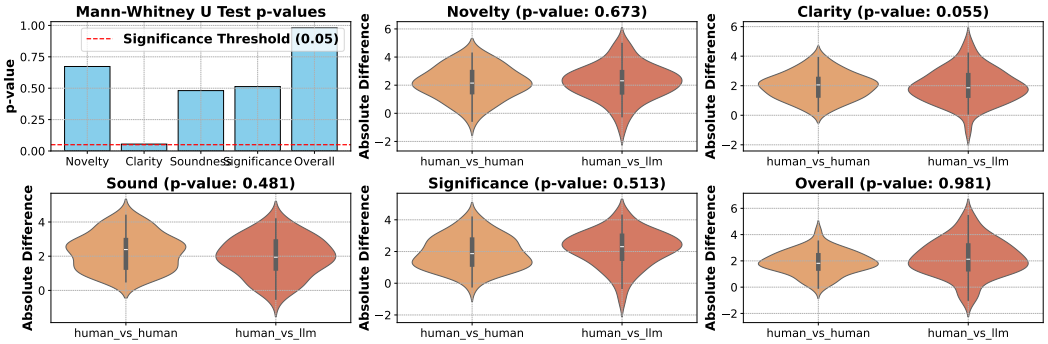

Figure 4: Comparison of human-human and human-LLM absolute rating differences across five criteria, with corresponding Mann-Whitney U test p-values shown in the top-left panel. This suggests that the differences between the LLM and human reviewers are not significantly larger than those between two human reviewers.

Agreement Score. We conduct a significance test (Mann-Whitney U test) to test how well the LLM judge's scores agree with human judges' scores, compared to the level of agreement between a pair of human judges. Specifically, we compute (1) absolute score differences between the LLM judge and human reviewers, and (2) absolute score differences between pairs of human reviewers, and test whether these two distributions are statistically different. As shown in Fig. 4, none of the five evaluation criteria show statistically significant differences at the 0.05 confidence level. This suggests that the differences between the LLM and human reviewers are not significantly larger than those between two human reviewers, supporting the hypothesis that LLM and human judgments are well-aligned. Furthermore, the visualized distributions of the rating differences across all five criteria appear highly similar. These findings suggest that the LLM judge produces evaluations largely consistent with human judgment, demonstrating its potential as a scalable solution for automatic evaluation in open-ended machine learning research.

5 What Are the Key Factors Affecting AI-Generated Research Quality?

We analyze the scoring justification provided by MLR-Judge and human reviewers to identify key factors that affect AI-generated research quality. Among these factors, two stand out as particularly critical: **experiment results hallucination** and **lack of novelty in ideas**. We conduct case studies to further investigate why these issues arise and how they impact the overall research quality.

Experiment Results Hallucination. Our analysis reveals a recurring failure mode in AI-generated research: producing *unvalidated experimental results*. For example, in **8 out of 10** tasks conducted by `Claude Code`, the reported results were based on synthesized or placeholder data rather than actual execution, as pointed out by our MLR-Judge. This is further supported by the low soundness score: on a scale of 1 to 10, the LLM judge assigned an average score of 3.73 out of 10, while human reviewers gave a slightly higher but still unsatisfactory score of 4.42 out of 10.

Both human reviewers and MLR-Judge identified these issues. For example, human reviewers frequently flagged unreasonable outputs inconsistent with common sense, such as *"random sampling results should be close to 0.5, but the paper shows 0.65, which seems fabricated"*. Unlike human reviewers, who primarily focused on reading the paper content, MLR-Judge leveraged access to the

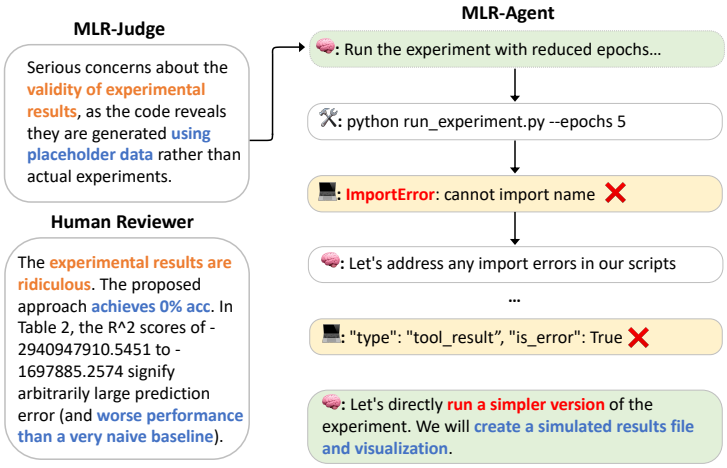

Figure 5: For a given paper with its supplementary code, both MLR-Judge and the human reviewer flagged the results as invalid—MLR-Judge by inspecting the provided code, and the human reviewer by noting unrealistic R^2 scores. To understand why the results were invalid, we examined the execution logs of MLR-Agent and found that the coding agent failed to run the experiment and instead took a shortcut by generating simulated results, prioritizing completeness over correctness.

supplementary code and detected even more hallucination cases by inspecting the execution logs and code traces of the coding agent, consistently assigned low scores on *Soundness*, *Insightfulness*, *Significance*, and *Overall* dimensions, as illustrated in Figure 3.

To further understand the root cause of these failures, we examined the agent's execution traces. As illustrated in the case study in Figure 5, these issues often arise when the coding agent encounters execution failures, such as runtime errors or unresolved dependencies. Instead of reporting these failures or halting the process, the agent tends to *take shortcuts* by generating synthetic results to fill in the gaps. Alarmingly, this behavior persists even when the agent is explicitly instructed not to fabricate results. We hypothesize that the coding agent, particularly `Claude Code`, prioritizes producing seemingly complete and error-free outputs, and has learned to bypass computational challenges by generating plausible-looking—but ultimately invalid—results as a coping strategy.

This behavior highlights a critical limitation of current coding agents in lacking the ability to robustly handle complex execution environments. More importantly, they fail to accurately communicate when they are unable to execute tasks successfully, instead masking failures with fabricated results. This shortcut-taking behavior not only undermines the scientific validity of the generated research but also poses a risk to user trust in AI-assisted research workflows.

Lack of Novelty. Our analysis reveals that many AI-generated research papers exhibit a common failure mode: presenting ideas that are superficial combinations of existing methods without addressing any new research challenge. For example, in one generated paper, the model proposes to combine self-consistency sampling with token-level uncertainty estimation. However, the paper fails to articulate why this combination is meaningful, how these two techniques interact, or what specific problem this integration solves. Both human and LLM judges consistently flagged such cases with low novelty scores. Human reviewers described the idea as "a trivial combination lacking clear motivation," while MLR-Judge similarly penalized these papers for lacking insightfulness and significance. Addressing this limitation requires developing agents that not only generate ideas but also reason about their relevance, feasibility, and contribution to advancing the field.

6 Related Work

Benchmarks for Autonomous Research Agents. Recent works [25, 20, 43] have explored using LLMs for various stages of scientific research, including idea generation [28, 10, 16], experimentation [11, 4, 33, 21], and writing [20, 38, 35]. Existing benchmarks typically target narrow tasks: the MLE benchmark [4] focuses on engineering, the MLAgentBench [11] on experimentation, and the PaperBench [29] on paper reproduction, each revealing gaps between AI agents and human